Online supplementary table S4. Characteristics of primary studies (shading indicates reason for exclusion from pooled analysis; yellow indicates exclusion in favour of other analysis of same cohort of greater duration or geographic scope; blue indicates results were provided only from multi-pollutant models [4], or there were too few studies of the outcome [16, 17, 26, 36], green indicates exclusion from pooled analysis in favour of ESCAPE[a] multi-cohort analysis)

Study	Cohort[a]	Location	Country/ Region	Start	End	N	Exposure	Minimum Age	Cause of Death[b]	Mean NO_2 (ppb)[c]
Crouse 2015 [1]	CanCHEC 1991	Canada	Canada	1991	2006	2,521,525	model	25	AC/NC, CEV, COPD, CV, IHD, LC, RES	11.6
Crouse 2015 [2]	CanCHEC 1991	10 Canadian cities	Canada	1991	2006	735,590	model	25	AC/NC, CEV, CV, IHD, RES	25.2
Weichenthal 2017 [3]	CanCHEC 2001	Canada	Canada	2001	2011	2,448,500	model	25	AC/NC, CV, RES	11.5
Villeneuve 2013 [4]	OTCS	Toronto, Canada	Canada	1982	2004	58,760	model	35	AC/NC, C, CV, LC, RES	28.9
Chen 2013 [5]	OTCS	Toronto, Hamilton, Windsor, Canada	Canada	1982	2004	205,440	model	35	CEV, CV, IHD	23.3
Jerrett 2009 [6]	Toronto Western Hospital	Toronto, Canada	Canada	1992	2002	2,360	model	NA	AC/NC, CV, RES	23.1
Gan 2011 [7]	Vancouver CHD	Vancouver, Canada	Canada	1999	2002	452,735	model	45	IHD	17.1
Gan 2013 [8]	Vancouver COPD	Vancouver, Canada	Canada	1999	2002	467,994	model	45	COPD	17.1
Abbey 1999 [9]	AHSMOG	California	United States	1977	1992	6,338	monitor	27	AC/NC, CPD, LC, RES	36.8
Chen 2005 [10]	AHSMOG	San Francisco, South Coast, San Diego	United States	1977	1998	3,239	monitor	25	IHD	34.9
Eckel 2016 [11]	California cancer	California	United States	1988	2009	352,053	monitor	NA	AC/NC, LC	21.9
Hartiala 2016 [12]	Cleveland Clinic GeneBank	Ohio	United States	2001	2010	4,363	monitor	NA	AC/NC	18.2
Jerrett 2013 [13]	CPS II	California	United States	1982	2000	73,711	model	30	AC/NC, CEV, CV, IHD, LC, RES	12.3
Pope 2002 [14]	CPS II	United States	United States	1982	1998	500,000	monitor	30	AC/NC, CPD, LC	21.4
Krewski 2009 [15]	CPS II	United States	United States	1982	2000	406,917	monitor	30	AC/NC, CPD, IHD, LC	27.9
McKean-Cowdin 2009 [16]	CPS II	United States	United States	1982	2000	527,123	monitor	30	BC	21.3
Turner 2017 [17]	CPS II	United States	United States	1982	2004	623,048	model	30	CC	11.6
Turner 2016 [18]	CPS II	United States	United States	1982	2004	669,046	model	30	AC/NC, CEV, COPD, CV, IHD, LC, RES	11.6
Lipsett 2011 [19]	CTS	California	United States	1996	2005	101,784	monitor	30	AC/NC, CEV, CV, IHD, LC, RES	33.6
Krewski 2000 [20]	Harvard Six Cities	6 US cities	United States	1974	1989	8,111	monitor	25	AC/NC, CPD, LC	15.0
Eum 2019 [21]	Medicare	United States	United States	2000	2008	14,100,000	monitor	65	AC/NC, CEV, COPD, CV, HF, IHD, LC, PN, RES	14.8
Lefler 2019 [22]	NHIS	United States	United States	1987	2015	635,539	model	18	AC/NC,CPD	10.7
Hart 2013 [23]	NHS	United States	United States	1988	2008	93,807	model	30	AC/NC	13.9
Lim 2019 [24]	NIH-AARP	6 US states	United States	1995	2011	548,845	model	50	CA,CEV,CV,IHD	13.3
Lim 2019 [25]	NIH-AARP	6 US states	United States	1995	2011	548,780	model	50	AC/NC,CA,CEV,COPD,	11.1

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Study	Cohort[a]	Location	Country/ Region	Start	End	N	Exposure	Minimum Age	Cause of Death[b]	Mean NO_2 (ppb)[c]
									CV,HF,IHD,LC,PN,RES	
Lim 2018 [26]	NIH-AARP	6 US states	United States	1995	2011	549,735	model	50	DB	12.2
Hart 2011 [27]	TrIPS	United States	United States	1985	2000	39,948	model	NA	AC/NC, COPD, CV, IHD, LC, RES	14.2
Lipfert 2006 [28]	WU/EPRI Veterans	United States	United States	1997	2001	28,635	monitor	NA	AC/NC	13.6
Lipfert 2006 [29]	WU/EPRI Veterans	United States	United States	1989	2001	41,598	monitor	NA	AC/NC	21.5
Lipfert 2009 [30]	WU/EPRI Veterans	United States	United States	1976	2001	67,938	model	NA	AC/NC	11.1
Lipfert 2019 [31]	WU/EPRI Veterans	United States	United States	1976	2001	67,938	monitor	NA	AC/NC	27.1
Lipfert 2018 [32]	WU/EPRI Veterans	United States	United States	1976	2001	67,938	monitor	NA	AC/NC	27.0
Bauleo 2019 [33]	Civitavecchia	Civitavecchia, Italy	Europe	1996	2013	71,362	model	18	AC/NC,CD,CEV,COPD, CV,IHD,LC,ND,RES	1.8
Carey 2013 [34]	CPRD	England	Europe	2003	2007	830,429	model	40	AC/NC, CV, LC, RES	12.0
Sifaki-Pistolla 2017 [35]	Crete lung cancer	Crete, Greece	Europe	1992	2014	5,057	model	NA	LC	9.0
Raaschou-Nielsen 2012 [36]	DDCH	Copenhagen and Aarhus, Denmark	Europe	1993	2009	52,061	model	50	DB	9.0
Raaschou-Nielsen 2012 [37]	DDCH	Copenhagen and Aarhus, Denmark	Europe	1993	2009	52,061	model	50	AC/NC, CEV, CV, IHD	9.0
Hvidtfeldt 2019 [38]	DDCH	Copenhagen and Aarhus, Denmark	Europe	1993	2015	49,564	model	50	AC/NC,CV,LC,RES[d]	14.0
Sørensen 2014 [39]	DDCH	Copenhagen and Aarhus, Denmark	Europe	1993	2009	51,569	model	50	CEV	9.7
Andersen 2012 [40]	DDCH	Copenhagen and Aarhus, Denmark	Europe	1993	2006	52,215	model	50	CEV	10.0
Fischer 2015 [41]	DUELS	Netherlands	Europe	2004	2011	7,218,363	model	30	AC/NC, CV, LC, RES	17.0
Dimakopoulou 2014 [42]	ESCAPE	Europe	Europe	1985	various	307,553	model	NA	RES	NA
Beelen 2014 [43]	ESCAPE	Europe	Europe	1985	various	367,251	model	NA	AC/NC	NA
Beelen 2014 [44]	ESCAPE	Europe	Europe	1985	various	367,383	model	NA	CEV, CV, IHD	NA
Bentayeb 2015 [45]	GAZEL	France	Europe	1989	2013	20,327	model	NA	AC/NC, CV, RES	14.9
Tonne 2013 [46]	MINAP	England and Wales	Europe	2004	2010	154,204	model	25	AC/NC	10.0
Tonne 2016 [47]	MINAP	London, United Kingdom	Europe	2003	2010	18,138	model	25	AC/NC	19.7
Stockfelt 2015 [48]	MPPS	Gothenburg, Sweden	Europe	1970	2007	7,494	model	NA	AC/NC, CEV, CV, IHD, RES	12.7

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Study	Cohort[a]	Location	Country/ Region	Start	End	N	Exposure	Minimum Age	Cause of Death[b]	Mean NO_2 (ppb)[c]
Dehbi 2017 [49]	NSHD, SABRE	London, United Kingdom	Europe	1989	2015	7,529	model	40	CV	23.7
Beelen 2008 [50]	NLCS	Netherlands	Europe	1986	1996	117,528	monitor	55	CD, CEV, CV, HF, IHD	NA
Beelen 2008 [51]	NLCS	Netherlands	Europe	1986	1996	117,528	monitor	55	AC/NC, CV, LC, RES	19.6
Brunekreef 2009 [52]	NLCS	Netherlands	Europe	1986	1996	105,296	model	55	AC/NC, CV, LC, RES	20.5
Hoek 2002 [53]	NLCS	Netherlands	Europe	1986	1994	4,492	monitor	55	AC/NC, CPD, LC	19.5
Nafstad 2004 [54]	Norwegian men	Oslo, Norway	Europe	1972	1998	15,966	model	40	AC/NC, CEV, IHD, LC, RES	4.4
Naess 2007 [55]	Oslo	Oslo, Norway	Europe	1992	1998	143,842	model	51	COPD, CV, LC	20.7
Filleul 2005 [56]	PAARC	7 French cities	Europe	1974	2001	14,284	monitor	25	AC/NC, CPD, LC	NA
Klompmaker 2020 [57]	PHM	Netherlands	Europe	2013	2017	339,633	model	30	AC/NC,CV,LC,ND,RES	12.9
Cesaroni 2012 [58]	RoLS	Rome, Italy	Europe	2001	2006	684,000	model	45	AC/NC	24.1
Cesaroni 2013 [59]	RoLS	Rome, Italy	Europe	2001	2010	1,265,058	model	30	AC/NC, CEV, CV, IHD, LC, RES	23.2
Rosenlund 2008 [60]	Rome	Rome, Italy	Europe	1998	2005	6,513	model	35	AC	NA
Schikowski 2007 [61]	SALIA	North Rhine-Westphalia, Germany	Europe	1985	2003	4,750	monitor	55	CV	20.7
Gehring 2006 [62]	SALIA	North Rhine-Westphalia, Germany	Europe	1985	2003	4,752	monitor	50	AC/NC, CPD	20.7
Heinrich 2013 [63]	SALIA	North Rhine-Westphalia, Germany	Europe	1985	2008	4,752	monitor	55	AC/NC, CPD, LC, RES	20.7
Nieuwenhuijsen 2018 [64]	SIDIAP	Barcelona, Spain	Europe	2010	2014	792,649	model	18	AC	
Maheswaran 2010 [65]	SLSR	South London	Europe	1995	2006	3,320	model	NA	AC/NC, CEV	21.8
Desikan et al. 2016 [66]	SLSR	South London	Europe	1995	2012	1,800	model	NA	AC/NC	23.7
Heritier 2019 [67]	SNC	Switzerland	Europe	2000	2008	4,404,046	model	30	MI	14.7
Cao 2011 [68]	CNHS	China	Other	1991	2000	70,947	monitor	40	AC/NC, CV, LC, RES	15.1
Dirgawati 2019 [69]	HIMS	Perth, Australia	Other	1996	2012	11,483	model	65	AC/NC,CEV	7.1
Yang 2018 [70]	HKEHC	Hong Kong	Other	1998	2011	61,386	model	65	AC/NC,CEV,COPD,CV, IHD,PN,RES	53.9
Barratt 2018 [71]	HKEHC	Hong Kong	Other	1998	2014	60,548	model	65	AC/NC,CEV,COPD,CV, IHD,PN,RES	30.8
Kim 2017 [72]	NHIS-NSC	Seoul, Korea	Other	2007	2013	136,094	monitor	18	AC/NC, CV, MI	34.5

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Study	Cohort[a]	Location	Country/ Region	Start	End	N	Exposure	Minimum Age	Cause of Death[b]	Mean NO_2 (ppb)[c]
Chen 2016 [73]	Northern Chinese	Tianjin, Shenyang, Taiyuan, Rizhao, China	Other	1998	2009	39,054	monitor	23	AC/NC, LC	21.6
Yorifuji 2010 [74]	SEC	Shizuoka, Japan	Other	1999	2006	13,444	model	65	AC/NC, CEV, COPD, CV, IHD, LC, RES	19.0
Yorifuji 2013 [75]	SEC	Shizuoka, Japan	Other	1999	2009	13,412	model	65	AC/NC, CEV, COPD, CPD, CV, IHD, LC, RES	11.7
Dong 2012 [76]	Shenyang	Shenyang, China	Other	1998	2009	9,941	monitor	25	RES	24.4
Zhang 2011 [77]	Shenyang	Shenyang, China	Other	1998	2009	9,941	monitor	25	CEV, CV	24.4
Tseng 2015 [78]	Taiwan civil servants	Taipei, Taiwan	Other	1989	2008	43,227	monitor	NA	CV	NA
Katanoda 2011 [79]	Three Prefecture	Miyagi, Aichi, Osaka, Japan	Other	1983	1995	63,520	monitor	NA	LC, RES	NA

[a] AHSMOG, Adventist Health and smog; CanCHEC, Canadian Census Health and Environment Cohort; Cleveland Clinic GeneBank study; CPS-II, Cancer Prevention Study-II; CNHS, China National Hypertension Survey; CPRD, Clinical Practice Research Datalink; CTS, California Teachers Study; DDCH, Danish Diet, Cancer and Health; DUELS, Dutch Environmental Longitudinal Study; ESCAPE, European Study of Cohorts for Air Pollution Effects; GAZEL, GAZ and ELectricité; HIMS, Health in Men Study; HKEHC, Hong Kong Elderly Health Centres; JPHC, Japan Public Health Centre; MINAP, Myocardial Ischaemia National Audit Project; MPPS, Multifactor Primary Prevention Study; NHIS, National Health Interview Survey; NHIS-NSC, National Health Insurance Service–National Sample Cohort; NIH-AARP, National Institutes of Health, American Association of Retired Persons; NSHD, National Survey of Health and Development; NLCS, Netherlands Cohort study on Diet and Cancer; NHS, Nurses Health Study; OTCS, Ontario Tax Cohort study; PAARC, Pollution Atmosphérique et Affections Respiratoires Chroniques; PHM, Public Health Monitor; RoLS, Rome Longitudinal Study; SEC, Shizuoka elderly cohort; SIDIAP, Sistema d'Informació pel Desenvolupament de la Investigació en Atenció Primària; SLSR, South London Stroke Register; SABRE, Southall And Brent Revisited; SALIA, Study on the influence of Air pollution on Lung function, Inflammation and Aging; SNC, Swiss National Cohort; TRiPS, the Trucking Industry Particle Study; WU/EPRI, Washington University/Electric Power Research Institute.

[b] AC/NC, All cause/Natural cause; BC, brain cancer; C, cardiac; CA, cardiac arrest; CC, colorectal cancer; CPD, cardiopulmonary disease; CV, cardiovascular; CEV, cerebrovascular; COPD, chronic obstructive pulmonary disease; DB, diabetes; IHD, ischemic heart disease; LC, lung cancer; MI, myocardial infarction; ND, neurologic disease; PN, pneumonia; RES, respiratory.

[c] In some cases estimated from median and/or nitrogen oxides.

[d] Only results for lung cancer were included in pooled analysis. Other outcomes captured through ESCAPE multi-cohort analysis.

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